

DPP No. # 2 (JEE-ADVANCED)

Total Marks : 31		Max. Time : 27 min.
Single choice Objective Questions ('-1' negative marking) Q. 1 to Q.3	(3 marks 3 min.)	[09, 09]
Multiple choice Objective Questions ('-1' negative marking) Q.4 to Q.6	(4 marks 3 min.)	[16, 12]
Subjective Questions ('-1' negative marking) Q.7 to Q.8	(3 marks 3 min.)	[06, 06]

Question No.	1	2	3	4	5	6	7	8	Total
Marks Obtained									

- Consider the function $f(x) = ({}^{x+1}C_{2x-8}) ({}^{2x-8}C_{x+1})$
Statement - 1 : Domain of $f(x)$ is singleton.
Statement - 2 : Range of $f(x)$ is singleton.
 (A) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is a correct explanation for STATEMENT-1
 (B) STATEMENT-1 is True, STATEMENT-2 is True ; STATEMENT-2 is **NOT** a correct explanation for STATEMENT-1
 (C) STATEMENT-1 is True, STATEMENT-2 is False
 (D) STATEMENT-1 is False, STATEMENT-2 is True
- Number of roots of equation $3^{|x|} - |2 - |x|| = 1$ is
 (A) 0 (B) 2 (C) 4 (D) 7
- Consider the functions $f_1(x) = 2\log_2 x$ and $f_2(x) = \log_2 x^2$. If D_1, D_2 and R_1, R_2 be the domains and ranges of the two functions respectively, then
 (A) $D_1 = D_2$ and $R_1 = R_2$ (B) $D_1 \subseteq D_2$ and $R_1 = R_2$
 (C) $D_1 = D_2$ and $R_1 \subset R_2$ (D) $D_1 \subseteq D_2$ and $R_1 \subset R_2$
- If (x_1, y_1) & (x_2, y_2) are the solutions of the equations, $\log_{225}(x) + \log_{64}(y) = 4$ and $\log_x(225) - \log_y(64) = 1$, then
 (A) $\log_{225}x_1 \cdot \log_{225}x_2 = 4$ (B) $\log_{225}x_1 + \log_{225}x_2 = 6$
 (C) $|\log_{64}y_1 - \log_{64}y_2| = 2\sqrt{5}$ (D) $\log_{30}(x_1x_2y_1y_2) = 12$
- For positive real numbers a ($a > 1$), let p_a and q_a be the maximum and minimum values, respectively, of $\log_a(x)$ for $a \leq x \leq 2a$. If $p_a - q_a = \frac{1}{2}$, then the value of a is not greater than
 (A) 3 (B) 4 (C) 5 (D) 6
- For the expression $(x \in \mathbb{R}) \frac{1}{\sin^6 x + \cos^6 x}$.
 (A) Maximum value is 4 (B) Minimum value is 1
 (C) Maximum value is 6 (D) Minimum value is $1/2$
- Determine the smallest positive value of x (in degrees) for which $\tan(x + 100^\circ) = \tan(x + 50^\circ) \tan x \tan(x - 50^\circ)$.
- If $\sin x + \sin y \geq \cos \alpha \cos x, \forall x \in \mathbb{R}$, then $\sin y + \cos \alpha$ is equal to

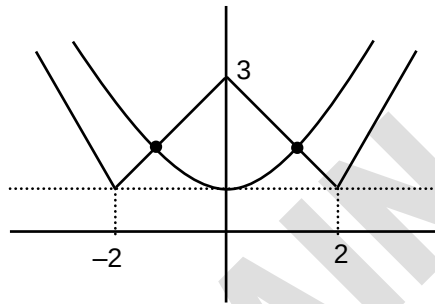
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1. (B) 2. (B) 3. (B) 4. (ABCD) 5. (BCD) 6. (AB)
 7. $x = 30^\circ$ 8. 1

Solution
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1. Defined for $x + 1 \geq 2x - 8$ and $2x - 8 \geq x + 1$
 hold when $x + 1 = 2x - 8 \Rightarrow x = 9$
 Domain is $\{9\}$ and Range is $({}^{10}C_{10}) ({}^{10}C_{10}) = \{1\}$

2. $3|x| = 1 + ||x| - 2|$



3. $D_1 = (0, \infty)$, $D_2 = (-\infty, 0) \cup (0, \infty) \Rightarrow D_1 \subseteq D_2$
 $R_1 = R_2 = (-\infty, \infty)$

4. $\frac{1}{\log_{225}^x} - \frac{1}{\log_{64}^y} = 1 \Rightarrow \log_{64}^y - \log_{225}^x = \log_{225}^x \cdot \log_{64}^y$

from equation (1) $4 - 2\log_{225}^x = \log_{225}^x (4 - \log_{225}^x)$

$$(\log_{225}^x)^2 - 6\log_{225}^x + 4 = 0$$

so $\log_{225}^{x_1} + \log_{225}^{x_2} = 6$, $\log_{225}^{x_1} \cdot \log_{225}^{x_2} = 4$

$$x_1 x_2 = (225)^6$$

Similarly

$$(\log_{64}^y)^2 - 2(\log_{64}^y) - 4 = 0$$

$$\log_{64}^{y_1} + \log_{64}^{y_2} = 2$$

$$\log_{64}^{y_1} \cdot \log_{64}^{y_2} = -4 \quad \therefore |\log_{64}^{y_1} - \log_{64}^{y_2}| = 2\sqrt{5}$$

$$\text{Also } y_1 y_2 = (64)^2$$

$$\text{so } x_1 x_2 y_1 y_2 = (225 \times 4)^6 = \log_{30}^{(30)^{12}} = 12$$

5. $P_a = \log_a^{2a} = \log_a^2 + \log_a^a = 1 + \log_a^2$

$$Q_a = \log_a^a = 1$$

$$P_a - Q_a = \log_a^2 = \frac{1}{2}$$

$$\sqrt{a} = 2 \Rightarrow a = 4$$

a is not greater than 4, 5, 6

6. $\frac{1}{(\sin^2 x + \cos^2 x)^3 - 3\sin^2 x \cos^2 x (\sin^2 x + \cos^2 x)} = \frac{1}{1 - \frac{3}{4}(\sin 2x)^2}$

$$= \max. = \frac{1}{1 - \frac{3}{4}(1)} \quad \min = \frac{1}{1 - \frac{3}{4}(0)} \quad \max = 4 \quad \min = 1$$

7. $\frac{\sin(x+100)\cos x}{\cos(x+100)\sin x} = \frac{\sin(x+50^\circ)\sin(x-50^\circ)}{\cos(x+50^\circ)\cos(x-50^\circ)}$

C – D Rule

$$\Rightarrow \frac{\sin(2x+100^\circ)}{\sin(100^\circ)} = \frac{\cos 100^\circ}{-\cos 2x}$$

$$\Rightarrow \sin(2x+100^\circ)\cos 2x = \sin 10^\circ \cos 10^\circ$$

$$\Rightarrow \sin(4x+100^\circ)\sin 100^\circ = \sin 20^\circ$$

$$\Rightarrow \sin(4x+100^\circ) = -\sin 40^\circ$$

$$\Rightarrow \sin(4x+100^\circ) = \sin 220^\circ$$

$$\Rightarrow x = 30^\circ$$

8. $\sin x + \sin y \geq \cos \alpha \cdot \cos x \quad \forall x \in \mathbb{R}$.

Let $x = -\frac{\pi}{2}$

$$\Rightarrow \sin y \geq 1 \Rightarrow (\sin y = 1) \Rightarrow 1 + \sin x \geq \cos \alpha \cos x \Rightarrow \cos \alpha \cos x - \sin x \leq 1$$

$$\Rightarrow \sqrt{\cos^2 \alpha + 1} \leq 1 \Rightarrow \cos \alpha = 0$$